# Automated Experiment Report

Run folder: run_20260418_235229

Prepared on 2026-04-19 12:55:58. This report was created automatically from saved run artifacts using deterministic Python rules only.

| | |
|---|---|
| **Model family**<br>Direct Hurdle Tabular | **Horizons**<br>24, 48 |
| **Platform**<br>Linux-6.6.113+-x86_64-with-glibc2.35 | **XGBoost device**<br>cuda |
| **Classifier configs**<br>2 | **Regressor configs**<br>2 |
| **Pairs per horizon**<br>4 | **Total fold fits**<br>16 |

## Executive summary

- The run archived 192 tracked artifacts.
- Best 24h pair: h24__clf_logen01__reg_xgbr01.
- Best 48h pair: h48__clf_xgbc01__reg_xgbr01.
- For horizon 24h, the winning pair was Logistic Enet + Xgb Regressor. Average county RMSE expected was 60.3997; Improved by 3.2120 (5.05%). Stage-1 ROC AUC was 0.8679, indicating strong discrimination.
- For horizon 48h, the winning pair was Xgb Classifier + Xgb Regressor. Average county RMSE expected was 60.0790; Improved by 4.1838 (6.51%). Stage-1 ROC AUC was 0.8821, indicating strong discrimination.
- Submission-style CSVs were exported from: Best validation candidate.
- Demo-only evaluation files are present. They are useful for sanity checks only and are not the hidden challenge evaluation.

## Configuration and modeling decisions

- Architecture: the run uses a two-stage direct hurdle pipeline ('direct_hurdle_tabular'). Stage 1 predicts whether outage is non-zero; Stage 2 predicts outage size conditional on positive outcomes only.
- Combination rule: the final expected outage equals stage-1 non-zero probability times the stage-2 positive prediction.
- Target transform: Stage 2 uses log1p on positive outages before fitting the regressor.
- Forecast horizons: 24h, 48h.
- Forecast-origin sampling: every 3 hours were sampled.
- Minimum training history before the first fold: 21 days (504 h).
- Maximum training history: capped at 28 days (672 h).

- Temporal validation design: 2 fold(s) per horizon were evaluated.
- Target lag features: 24, 48 hours.
- Target rolling windows: 6 hours.
- Weather lag features: 24, 48 hours.
- Weather rolling windows: 6 hours.
- Weather screening: up to 50 weather variables were allowed; 50 were retained in this run.
- Weather outlier handling: enabled with 'winsorize_iqr', so extreme weather values were clipped rather than turned into missing values.
- Execution hardware: GPU acceleration was available; XGBoost used device 'cuda'.
- Prediction export policy: submission CSVs were generated from Best validation candidate.

### Search space summary

| component | families | config_count |
|---|---|---|
| Classifier | logistic_enet, xgb_classifier | 2 |
| Regressor | elasticnet_regression, xgb_regressor | 2 |

## Run inventory

The table below shows the top-level artifact folders detected in the run. The checkpoints folder is runtime infrastructure and may be intentionally absent from shared GitHub uploads.

| subfolder | present | file_count | note |
|---|---|---|---|
| tables | Yes | 33 | |
| predictions | Yes | 20 | |
| models | Yes | 64 | |
| metrics | Yes | 44 | |
| plots | Yes | 17 | |
| checkpoints | Yes | 12 | Runtime-only resume artifacts; may be omitted from shared GitHub uploads. |

## Temporal evaluation design

The backtest table below maps the training and validation windows used for each horizon and fold.

| horizon | fold_id | val_origin_idx | val_origin_timestamp | train_origin_max_idx | train_origin_max_timestamp | validation_window_start | validation_window_end |
|---|---|---|---|---|---|---|---|
| 24 | 1 | 1,800 | 2023-06-15 | 1,776 | 2023-06-14 | 2023-06-15 01:00:00 | 2023-06-16 |
| 24 | 2 | 2,136 | 2023-06-29 | 2,112 | 2023-06-28 | 2023-06-29 01:00:00 | 2023-06-30 |
| 48 | 1 | 1,776 | 2023-06-14 | 1,728 | 2023-06-12 | 2023-06-14 01:00:00 | 2023-06-16 |
| 48 | 2 | 2,112 | 2023-06-28 | 2,064 | 2023-06-26 | 2023-06-28 01:00:00 | 2023-06-30 |

## Feature frame and data overview

| metric | value |
|---|---|
| rows_in_model_df | 179,363 |
| base_numeric_features | 274 |
| categorical_features | 1.0000 |
| feature_warmup_hours | 48 |
| max_horizon | 48 |
| mean_outage | 45.2483 |
| median_outage | 0.0000 |
| std_outage | 452.2684 |
| max_outage | 23,346 |
| zero_rate | 0.7047 |
| nonzero_rate | 0.2953 |

Selected weather features retained in this run: 50.

| feature | abs_corr_with_outage | signed_corr_with_outage |
|---|---|---|
| gh_4 | 0.1299 | 0.1299 |
| cape_1 | 0.0982 | 0.0982 |
| pwat | 0.0885 | 0.0885 |
| sh2 | 0.0832 | 0.0832 |
| mslma | 0.0758 | -0.0758 |
| sdlwrf | 0.0707 | 0.0707 |
| mcc | 0.0662 | 0.0662 |
| d2m | 0.0650 | 0.0650 |
| cape | 0.0641 | 0.0641 |
| lcc | 0.0554 | 0.0554 |
| tcc | 0.0539 | 0.0539 |
| pcdb | 0.0538 | 0.0538 |

# Candidate ranking by horizon

Rows are ranked within each horizon using average county RMSE expected first and overall RMSE expected second.

| rank | horizon | classifier_name | regressor_name | mean_avg_county_rmse_expected | mean_overall_rmse_expected | avg_county_gain_vs_zero | overall_gain_vs_zero | mean_stage1_roc_auc | mean_stage2_positive_rmse | pair_tag |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 24 | logistic_enet | xgb_regressor | 60.3997 | 148.5474 | 3.2120 | 15.0005 | 0.8679 | 266.3118 | h24__clf_logen01__reg_xgbr01 |
| 2 | 24 | xgb_classifier | xgb_regressor | 60.9420 | 153.5759 | 2.6697 | 9.9720 | 0.8815 | 266.3118 | h24__clf_xgbc01__reg_xgbr01 |
| 3 | 24 | logistic_enet | elasticnet_regression | 61.1048 | 148.6640 | 2.5069 | 14.8839 | 0.8679 | 266.9217 | h24__clf_logen01__reg_enet01 |

| rank | horizon | classifier_name | regressor_name | mean_avg_county_rmse_expected | mean_overall_rmse_expected | avg_county_gain_vs_zero | overall_gain_vs_zero | mean_stage1_roc_auc | mean_stage2_positive_rmse | pair_tag |
|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 24 | xgb_classifier | elasticnet_regression | 61.7150 | 153.9150 | 1.8967 | 9.6329 | 0.8815 | 266.9217 | h24__clf_xgbc01__reg_enet01 |
| 1 | 48 | xgb_classifier | xgb_regressor | 60.0790 | 143.8301 | 4.1838 | 16.9001 | 0.8821 | 257.4583 | h48__clf_xgbc01__reg_xgbr01 |
| 2 | 48 | logistic_enet | xgb_regressor | 60.1504 | 143.2717 | 4.1124 | 17.4585 | 0.8700 | 257.4583 | h48__clf_logen01__reg_xgbr01 |
| 3 | 48 | xgb_classifier | elasticnet_regression | 71.8474 | 225.4851 | -7.5846 | -64.7550 | 0.8821 | 413.4256 | h48__clf_xgbc01__reg_enet01 |
| 4 | 48 | logistic_enet | elasticnet_regression | 72.8874 | 230.5926 | -8.6246 | -69.8624 | 0.8700 | 413.4256 | h48__clf_logen01__reg_enet01 |

# Winning pairs by horizon

| horizon | classifier_name | regressor_name | mean_avg_county_rmse_expected | mean_overall_rmse_expected | avg_county_gain_vs_zero | overall_gain_vs_zero | mean_stage1_roc_auc | mean_stage1_f1 | mean_stage2_positive_rmse | pair_tag |
|---|---|---|---|---|---|---|---|---|---|---|
| 24 | logistic_enet | xgb_regressor | 60.3997 | 148.5474 | 3.2120 | 15.0005 | 0.8679 | 0.6809 | 266.3118 | h24__clf_logen01__reg_xgbr01 |
| 48 | xgb_classifier | xgb_regressor | 60.0790 | 143.8301 | 4.1838 | 16.9001 | 0.8821 | 0.7024 | 257.4583 | h48__clf_xgbc01__reg_xgbr01 |

Horizon 24h selected Logistic Enet + Xgb Regressor. Average county RMSE expected was 60.3997 versus a zero baseline of 63.6117. Improved by 3.2120 (5.05%).

Horizon 48h selected Xgb Classifier + Xgb Regressor. Average county RMSE expected was 60.0790 versus a zero baseline of 64.2628. Improved by 4.1838 (6.51%).

# Stage-specific summaries

## Stage 1 classifiers

| horizon | classifier_name | classifier_params | mean_stage1_roc_auc | mean_stage1_average_precision | mean_stage1_brier_score | mean_stage1_f1 |
|---|---|---|---|---|---|---|
| 24 | xgb_classifier | {"n_estimators": 150, "learning_rate": 0.08, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15} | 0.8815 | 0.8063 | 0.1382 | 0.5966 |
| 24 | logistic_enet | {"C": 0.5, "l1_ratio": 0.15, "class_weight": "balanced", "max_iter": 250} | 0.8679 | 0.7869 | 0.1437 | 0.6809 |
| 48 | xgb_classifier | {"n_estimators": 150, "learning_rate": 0.08, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15} | 0.8821 | 0.8166 | 0.1250 | 0.7024 |
| 48 | logistic_enet | {"C": 0.5, "l1_ratio": 0.15, "class_weight": "balanced", "max_iter": 250} | 0.8700 | 0.7899 | 0.1428 | 0.6930 |

## Stage 2 regressors

| horizon | regressor_name | regressor_params | mean_stage2_positive_rmse | mean_stage2_positive_mae | mean_stage2_positive_rows_train |
|---|---|---|---|---|---|
| 24 | xgb_regressor | {"n_estimators": 200, "learning_rate": 0.06, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15} | 266.3118 | 77.0833 | 134,083 |
| 24 | elasticnet_regression | {"alpha": 0.0005, "l1_ratio": 0.1, "max_iter": 4000} | 266.9217 | 82.6248 | 134,083 |
| 48 | xgb_regressor | {"n_estimators": 200, "learning_rate": 0.06, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15} | 257.4583 | 78.3202 | 266,357.5000 |

| horizon | regressor_name | regressor_params | mean_stage2_positive_rmse | mean_stage2_positive_mae | mean_stage2_positive_rows_train |
|---|---|---|---|---|---|
| 48 | elasticnet_regression | {"alpha": 0.0005, "l1_ratio": 0.1, "max_iter": 4000} | 413.4256 | 121.7192 | 266,357.5000 |

# Submission and refit artifacts

## Submission model summary

| horizon | prediction_model_source | pair_tag | fold_id | classifier_name | regressor_name | avg_county_rmse_expected | overall_rmse_expected |
|---|---|---|---|---|---|---|---|
| 24 | Best validation candidate | h24__clf_xgbc01__reg_xgbr01 | 1 | xgb_classifier | xgb_regressor | 49.7362 | 115.1983 |
| 48 | Best validation candidate | h48__clf_xgbc01__reg_enet01 | 1 | xgb_classifier | elasticnet_regression | 55.2359 | 119.3063 |

## Final refit summary

| horizon | bundle_source | pair_tag | classifier_name | regressor_name | mean_avg_county_rmse_expected | mean_overall_rmse_expected |
|---|---|---|---|---|---|---|
| 24 | Final refit | h24__clf_logen01__reg_xgbr01 | logistic_enet | xgb_regressor | 60.3997 | 148.5474 |
| 48 | Final refit | h48__clf_xgbc01__reg_xgbr01 | xgb_classifier | xgb_regressor | 60.0790 | 143.8301 |

# Prediction exports

| file | rows | prediction_column | min | max | mean | std |
|---|---|---|---|---|---|---|
| two_stage_pred_24h.csv | 1,992 | pred | 0.0801 | 192.4119 | 6.9295 | 24.2684 |
| two_stage_pred_48h.csv | 3,984 | pred | 0.2779 | 361.9636 | 11.5687 | 40.0813 |
| final_refit_pred_24h.csv | 1,992 | pred | 0.0086 | 550.1686 | 10.3747 | 41.5387 |
| final_refit_pred_48h.csv | 3,984 | pred | 0.0085 | 318.5515 | 6.7212 | 25.6096 |

The predictions folder also contains 16 pair_*_val_predictions.csv file(s) with row-level validation outputs for candidate analysis.

# Demo-only evaluation files

These metrics were computed against any demo labels bundled with the repository. They are useful for quick checks, but they are not the official hidden-test challenge results.

| file | overall_rmse | overall_mae | avg_county_rmse | n_rows | n_locations |
|---|---|---|---|---|---|
| demo_eval_24h.json | 25.1008 | 6.5752 | 6.7296 | 1,992 | 83 |
| demo_eval_48h.json | 41.5856 | 11.0965 | 11.2061 | 3,984 | 83 |
| demo_eval_final_refit_24h.json | 42.6869 | 9.9997 | 10.5363 | 1,992 | 83 |

| file | overall_rmse | overall_mae | avg_county_rmse | n_rows | n_locations |
|---|---|---|---|---|---|
| demo_eval_final_refit_48h.json | 26.3598 | 6.4585 | 6.7833 | 3,984 | 83 |

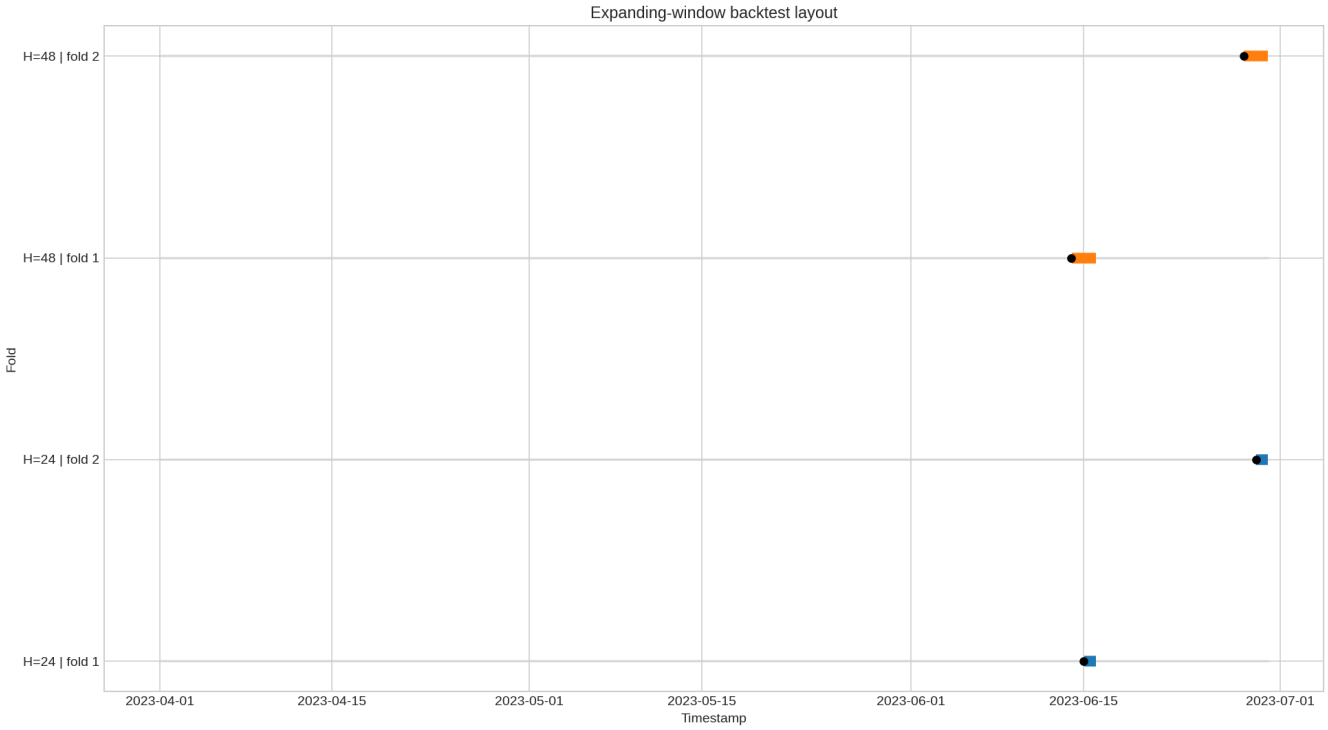
# Reusable model artifacts

| horizon | fold_id | pair_tag | classifier_name | regressor_name | two_stage_model_path | stage1_model_path | stage2_model_path |
|---|---|---|---|---|---|---|---|
| 24 | 1 | h24__clf_xgbc01__reg_xgbr01 | xgb_classifier | xgb_regressor | pair_h24__clf_xgbc01__reg_xgbr01_f01_two_stage.joblib | pair_h24__clf_xgbc01__reg_xgbr01_f01_stage1_classifier.joblib | pair_h24__clf_xgbc01__reg_xgbr01_f01_stage2_regressor.joblib |
| 48 | 1 | h48__clf_xgbc01__reg_enet01 | xgb_classifier | elasticnet_regression | pair_h48__clf_xgbc01__reg_enet01_f01_two_stage.joblib | pair_h48__clf_xgbc01__reg_enet01_f01_stage1_classifier.joblib | pair_h48__clf_xgbc01__reg_enet01_f01_stage2_regressor.joblib |

# Visual appendix

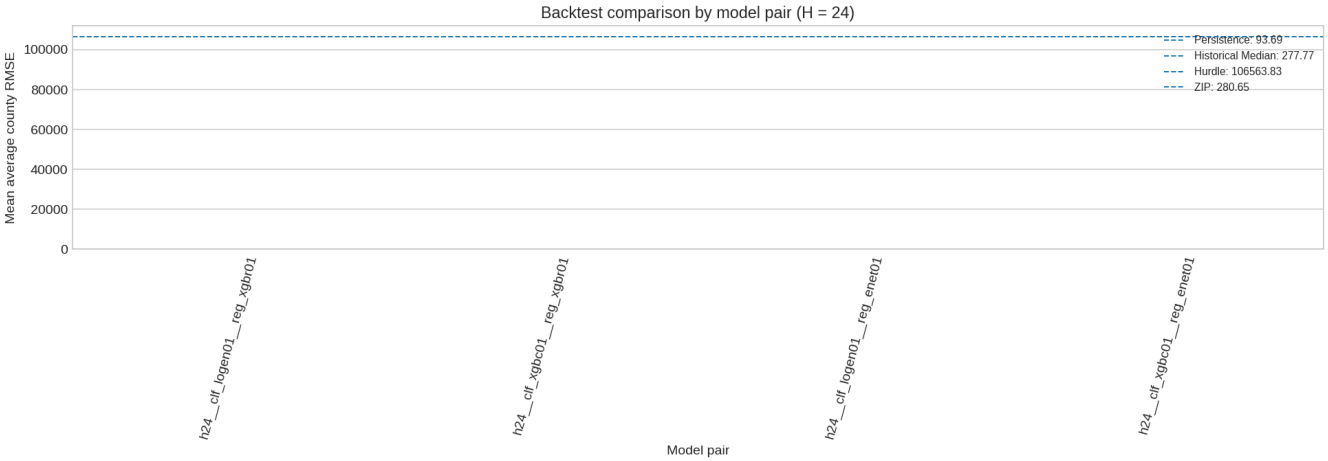
The following figures were read directly from the run's plots folder and inserted without GenAI interpretation.

## Backtest Layout



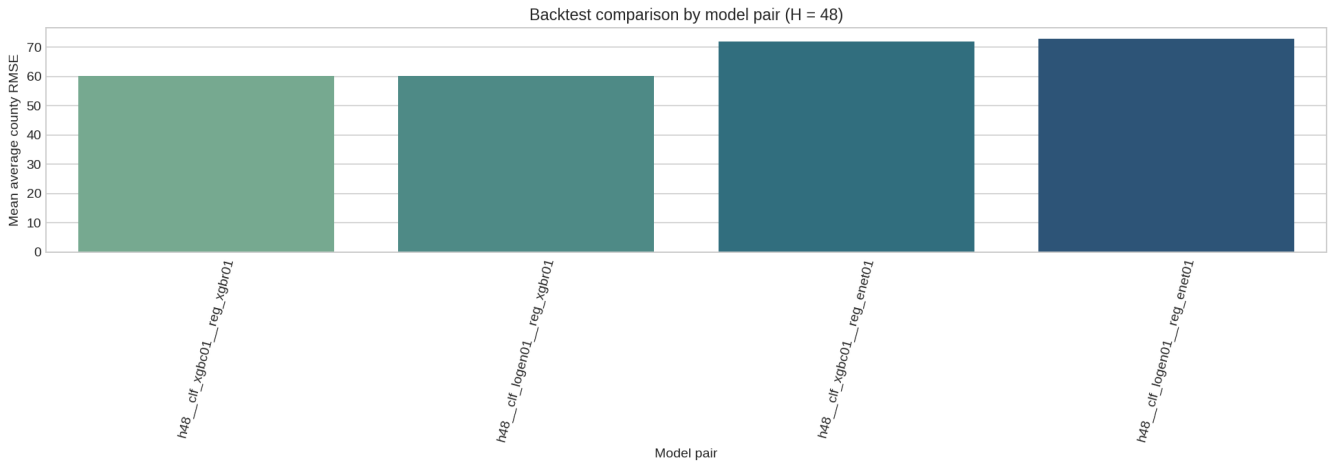

*backtest_layout.png*

## Pair Comparison 24H



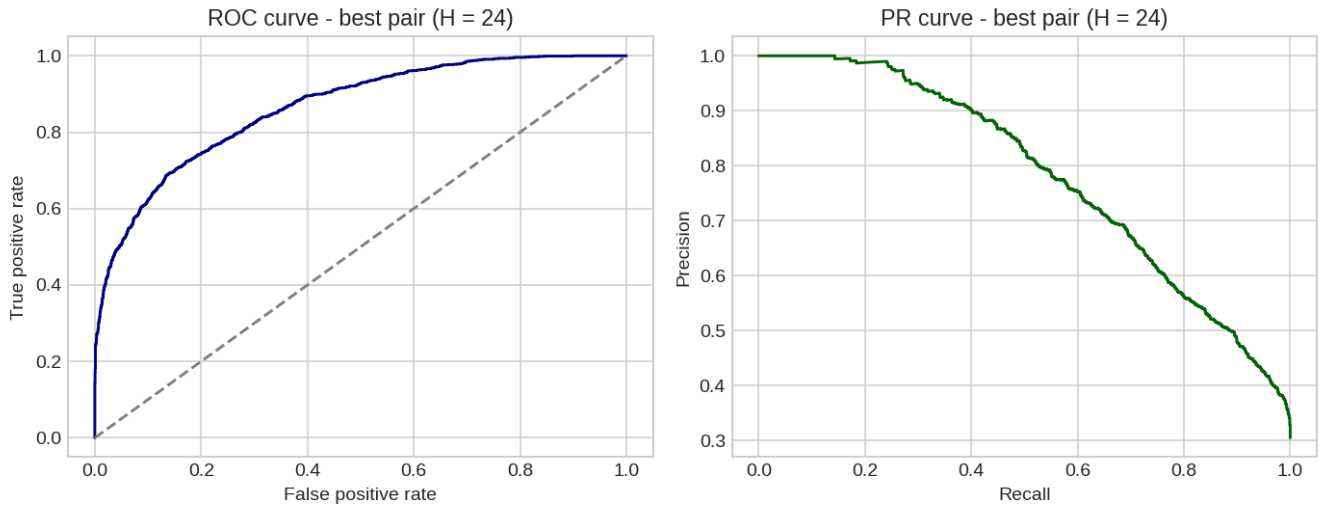

*pair_comparison_h24.png*

## Pair Comparison 48H



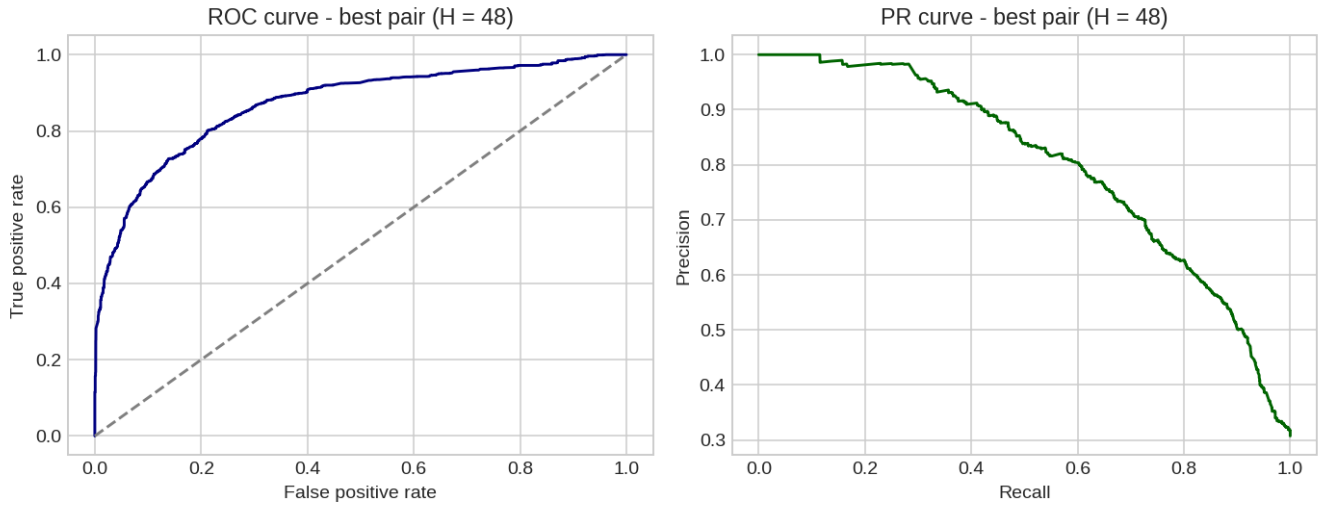

*pair_comparison_h48.png*

## Best Pair Curves 24H



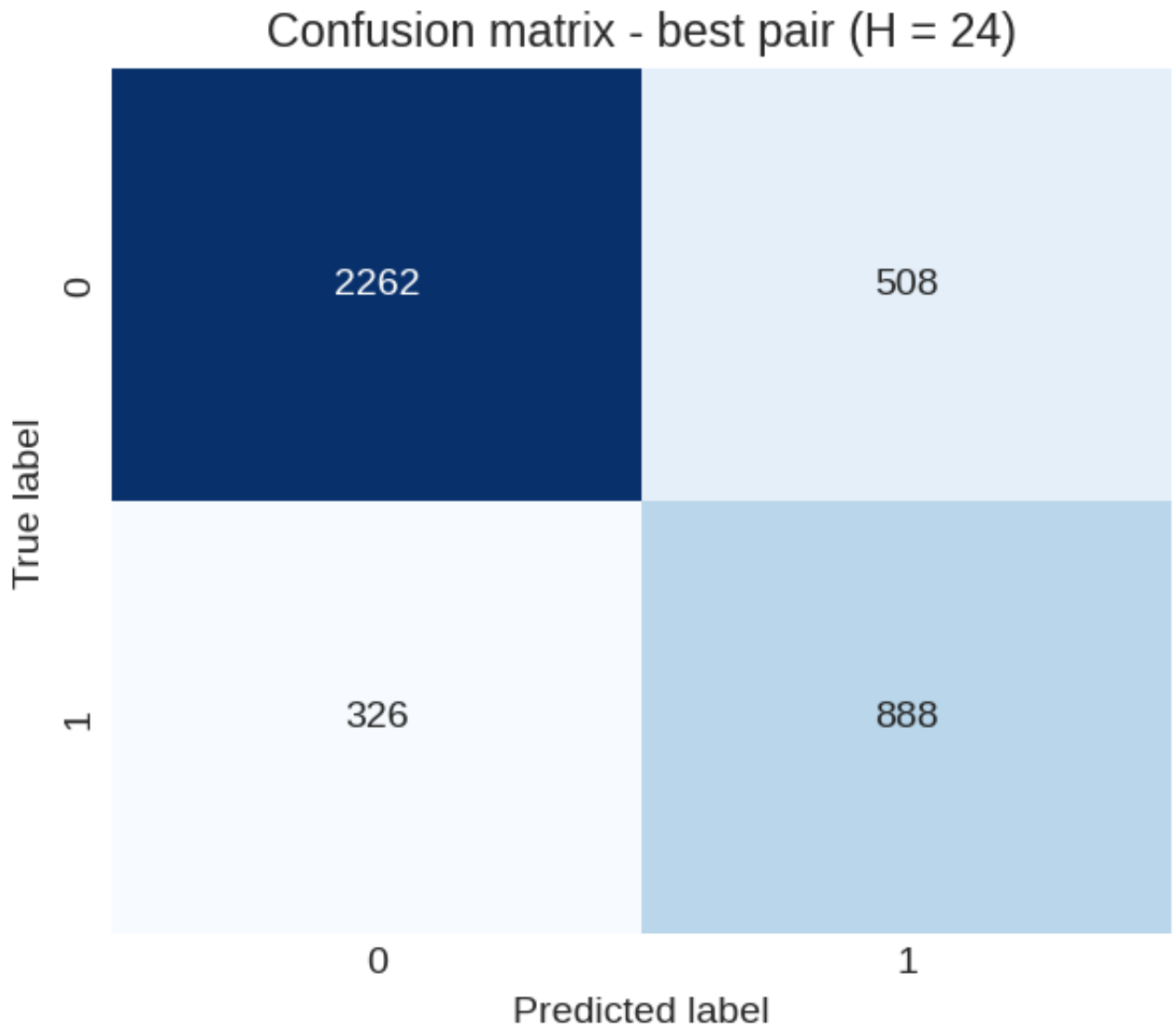

*best_pair_curves_h24.png*

## Best Pair Curves 48H



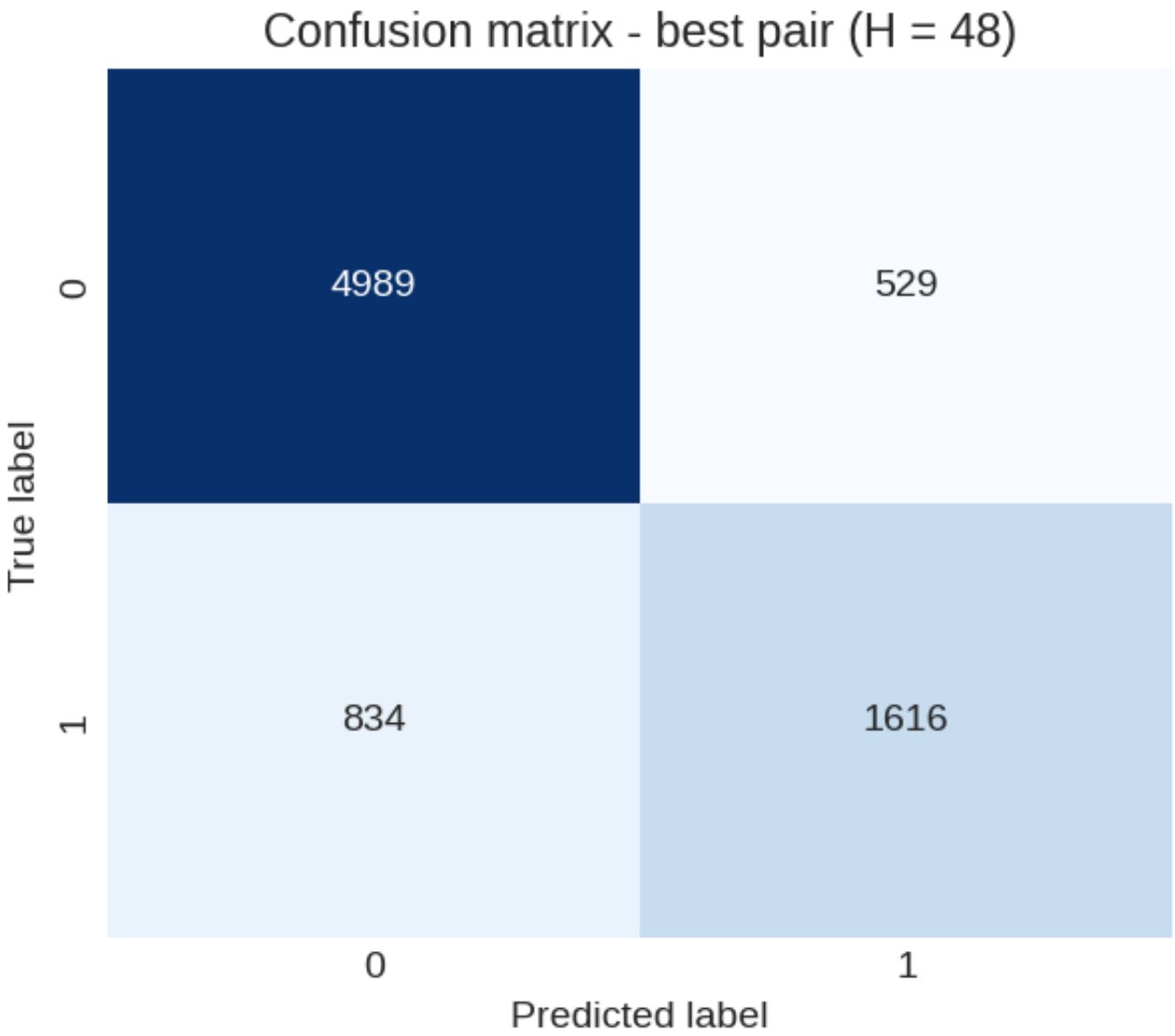

*best_pair_curves_h48.png*

## Best Pair Confusion 24H



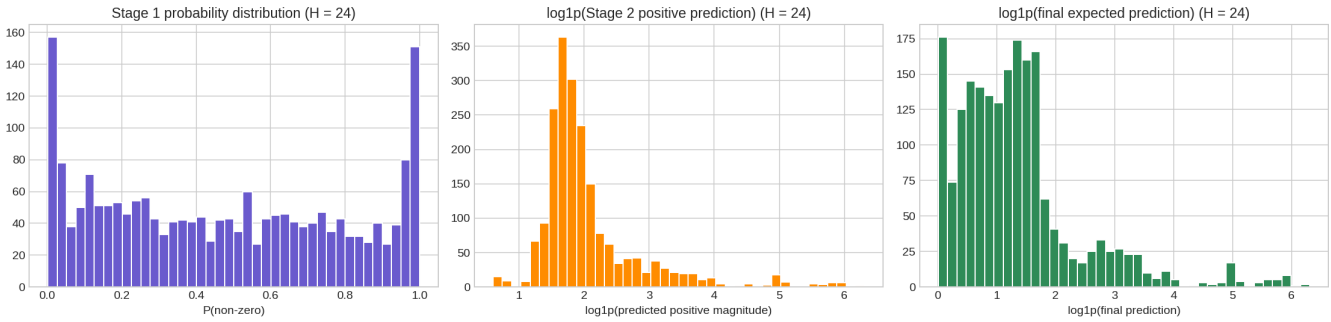

*best_pair_confusion_h24.png*

## Best Pair Confusion 48H



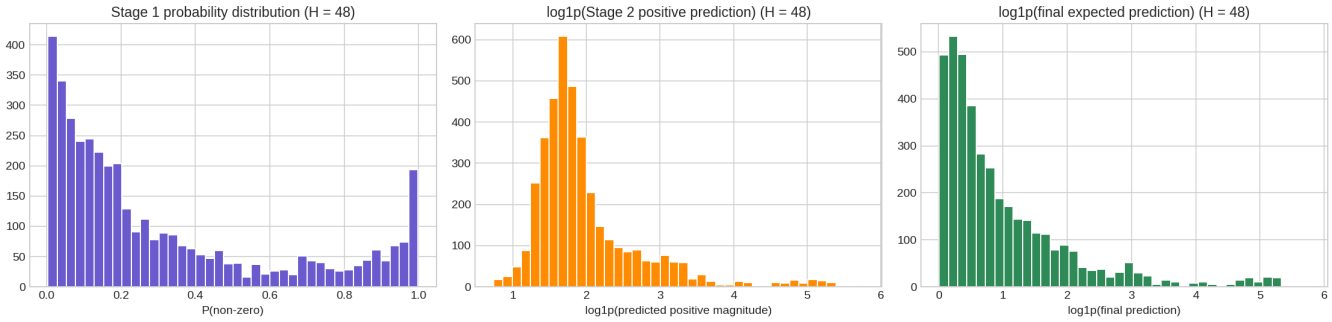

*best_pair_confusion_h48.png*

## Prediction Distributions 24H



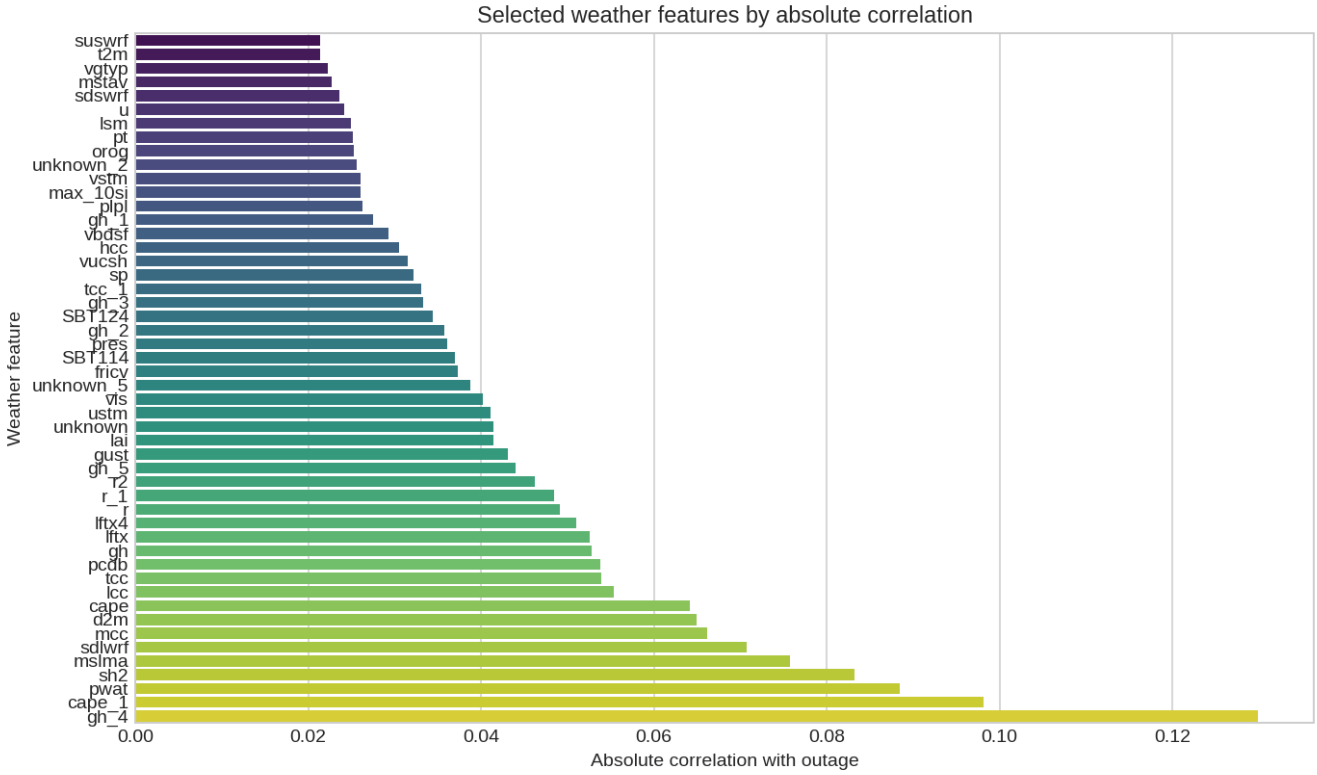

*prediction_distributions_h24.png*

## Prediction Distributions 48H



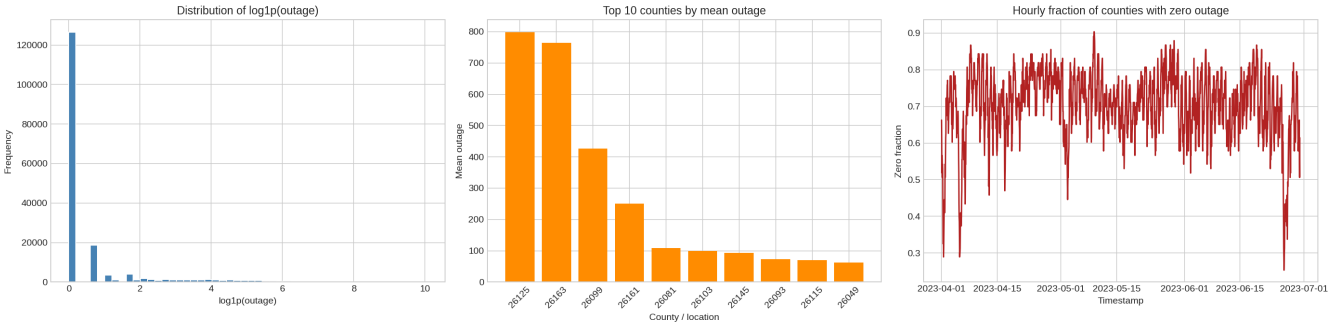

*prediction_distributions_h48.png*

## Selected Weather Features



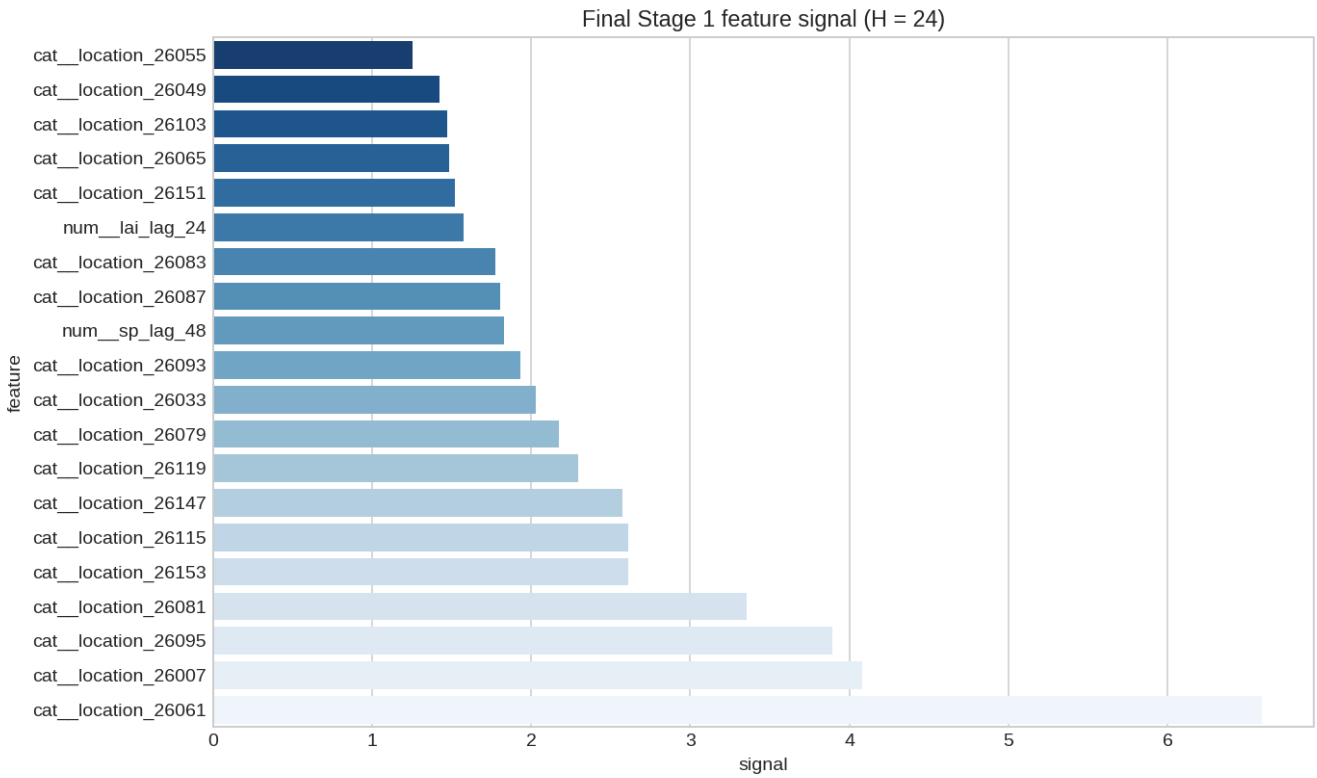

*selected_weather_features.png*

## Eda Overview



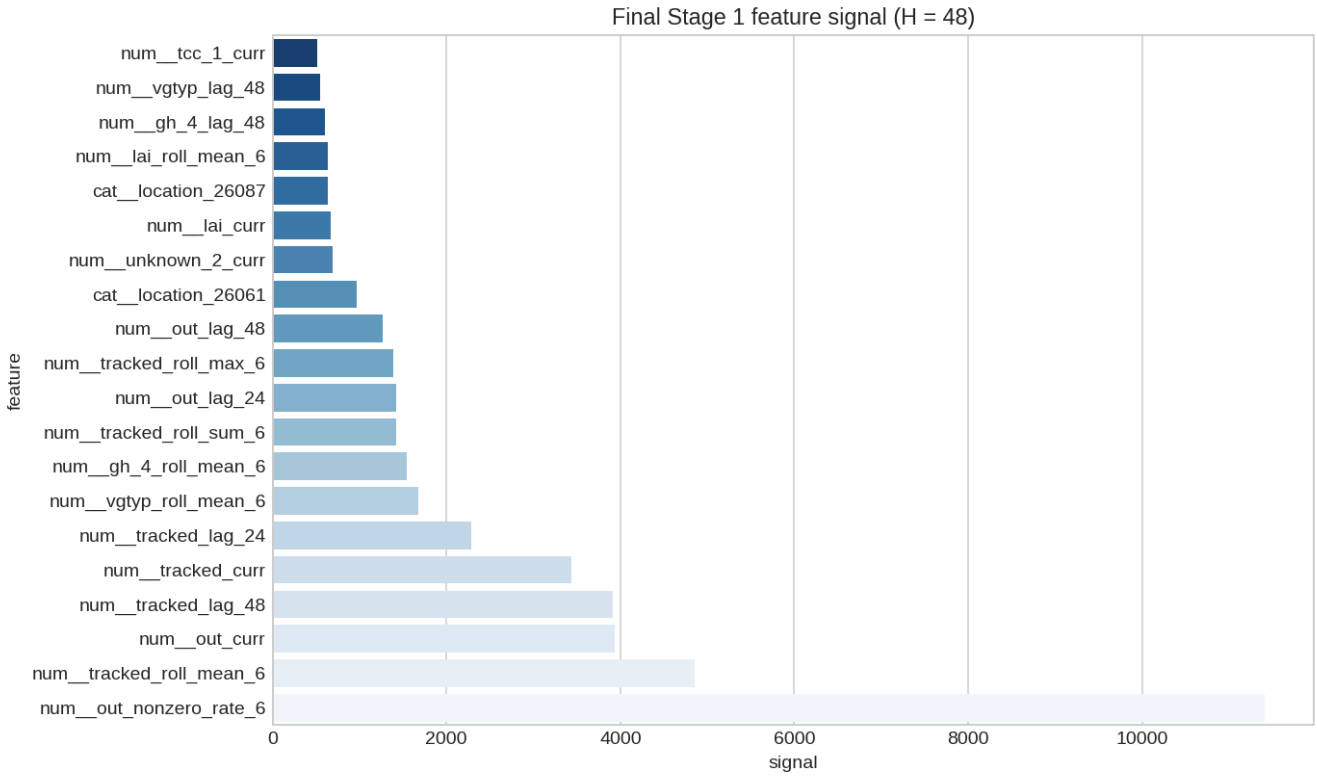

*eda_overview.png*

## Final Stage1 Feature Signal 24H



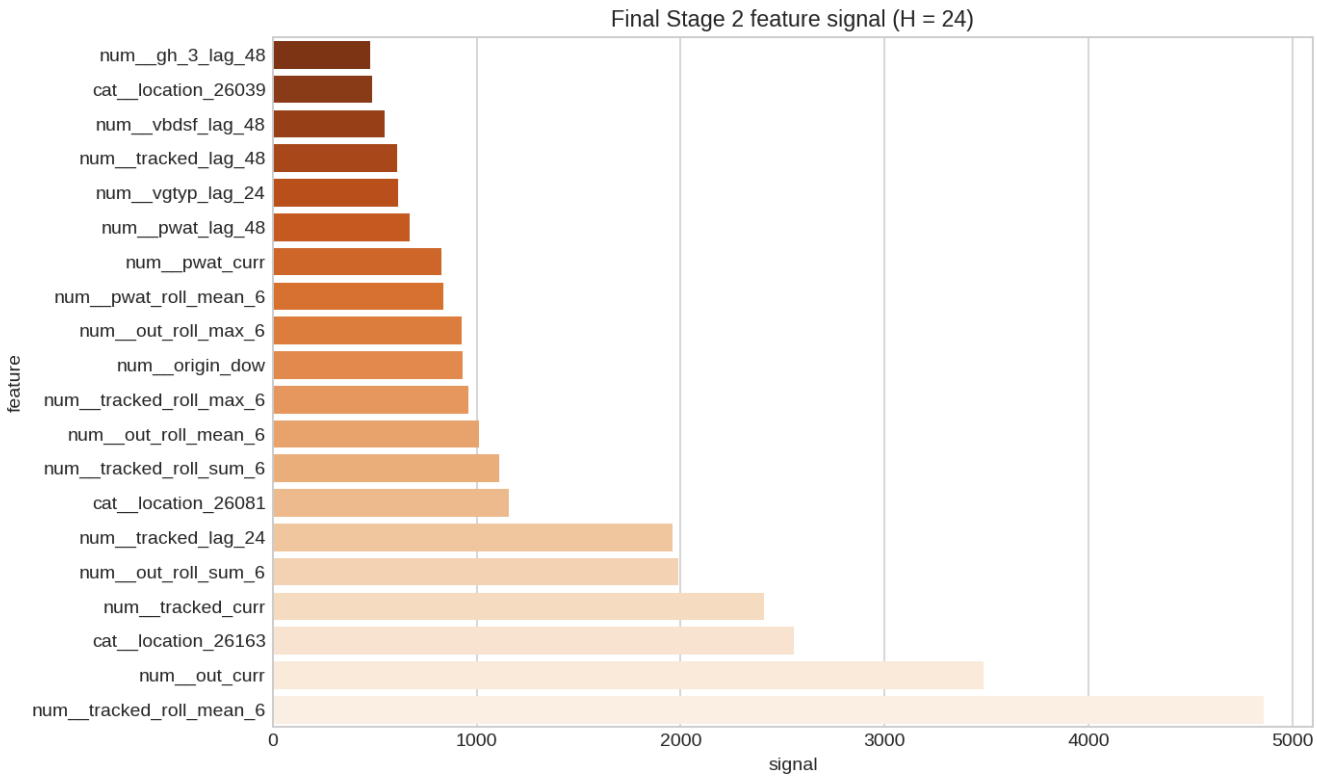

*final_stage1_feature_signal_h24.png*

## Final Stage1 Feature Signal 48H



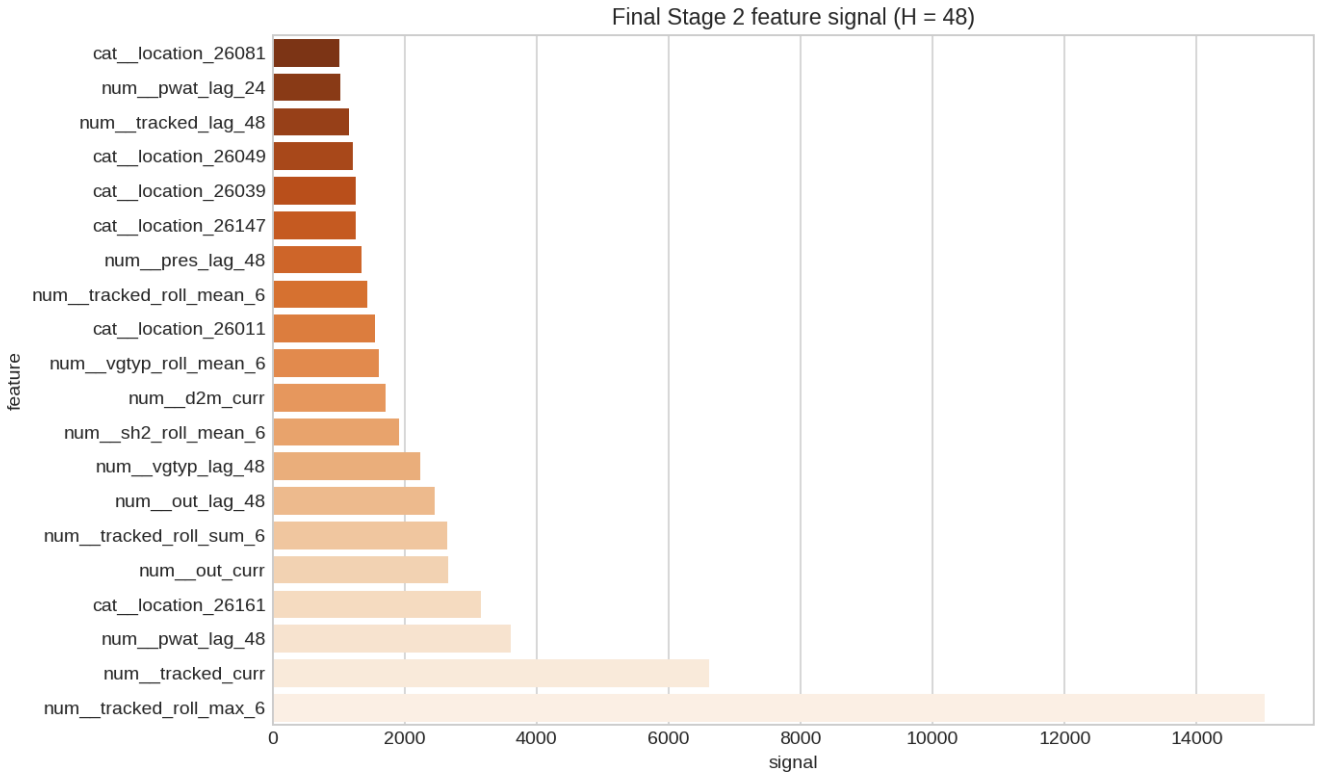

*final_stage1_feature_signal_h48.png*

## Final Stage2 Feature Signal 24H



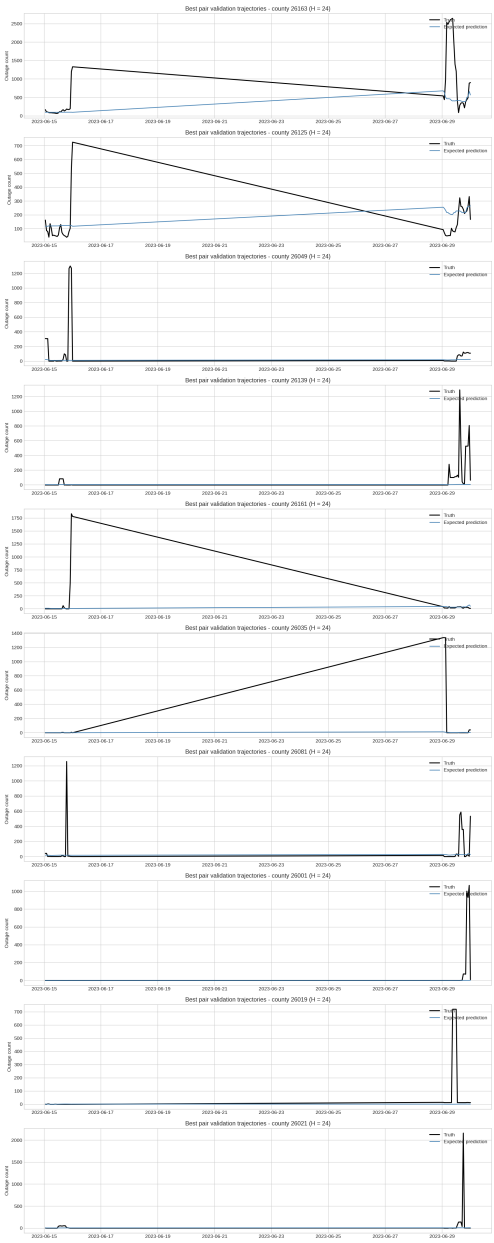

*final_stage2_feature_signal_h24.png*

## Final Stage2 Feature Signal 48H



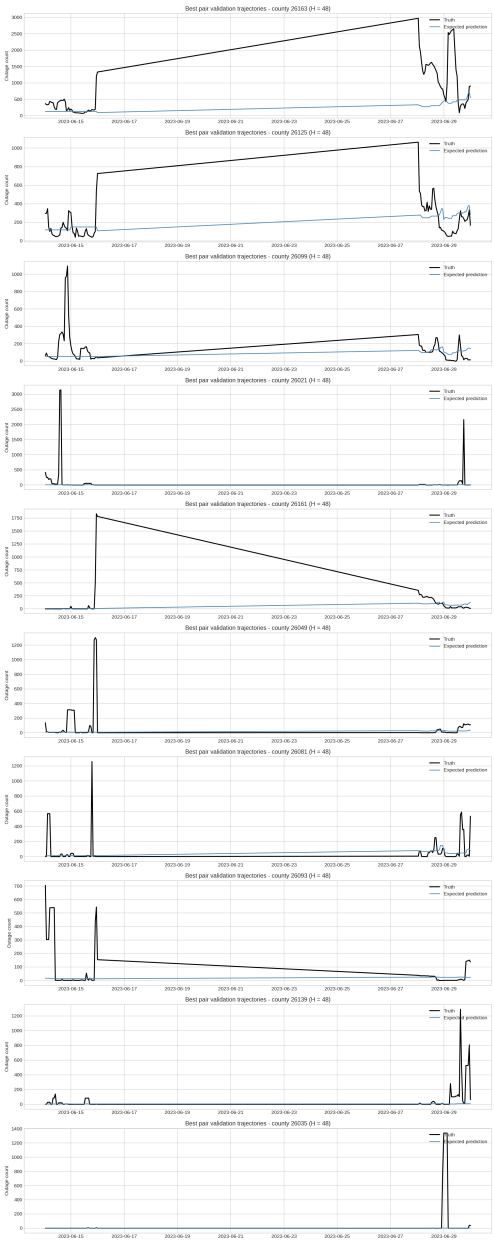

*final_stage2_feature_signal_h48.png*

## Best Pair Validation Trajectories 24H




*best_pair_validation_trajectories_h24.png*

## Best Pair Validation Trajectories 48H




*best_pair_validation_trajectories_h48.png*

# Folder guide

- tables/: aggregated summaries, rankings, split definitions, and run-level CSV records.
- predictions/: exported submission-style predictions and detailed candidate validation predictions.
- models/: reusable .joblib artifacts plus metadata JSON files for submission and refit bundles.
- metrics/: fold-level metrics, training histories, and optional demo evaluation JSON files.
- plots/: visual diagnostics such as split layouts, model comparisons, confusion matrices, and feature-signal plots.
- checkpoints/: runtime resume artifacts used during notebook execution. This folder may be intentionally absent from GitHub uploads.